

Final Experiment Programme

Design, Robustness and Domain of the Output-Only Transmissibility Inverse

SHARC Project Wednesday 30 September to Friday 9 October 2026 Thesis due 27 October

Rev. 7 (4 October, afternoon): V1 stage 0 and V1 results recorded; V1 clarifications, pre-run correction and files as built; truncation requirement for real data.

Rev. 6 (4 October): M1 result recorded (all four cases pass); M1 clarifications and files as built.

Rev. 5 (3 October): results of A1, R2 and R1 recorded; the four M1 scenarios fixed; V1's truncation "truth" to be revised first (A1 finding); amendments of 2–3 October listed; file table as built.

Rev. 4 (2 October): results of R4, D2, R3 and D1; new study D3; V2 promoted to SHOULD.

Why this plan exists. The supervisor meeting of 30 September removed the multi-board and wave-tank experiment from the critical path: the extra boards arrive on 7 October, tank access is unlikely, and a deployment would also need a waterproof enclosure and reliable inter-board communication. The remaining time goes to (i) documenting the large body of completed work and (ii) a systematic simulation programme around the frozen inverse. γ stays fixed, as a *known* per-scenario value from measured draught. Experiments stop on Friday 9 October; 10–27 October is thesis writing.

Research question for the remaining month. *Under what sensing, forcing, noise and floe conditions can the output-only transmissibility inverse reliably recover (β, R) , and how do we know when it cannot?*

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1 What is frozen, what is dropped		

1.1 Frozen pipeline

All studies vary the *conditions around* one fixed chain:

$$\text{EMM} \longrightarrow T_j(\omega) = \frac{H_j}{H_r} \longrightarrow \text{finite noisy record} \longrightarrow \hat{T}_j = \frac{\hat{S}_{jr}}{\hat{S}_{rr} - S_{nn}} \longrightarrow (\hat{\beta}, \hat{R}), \quad \gamma \text{ known.}$$

Frozen items: EMM formulation and truncation [50, 10, 10] (except where a study deliberately perturbs it), the Welch estimator (periodic Hann, $L = 2048$, 50% overlap, $N_{\text{eff}} = 59.7$), the every-second-bin grid, the 6×6 (in general $2(n_s - 1) \times 2(n_s - 1)$) block covariance, the weighted least-squares objective, the safeguarded Gauss–Newton solver and the pooled Hartlap-type jackknife.

Truncation status (A1, 3 October). [50, 10, 10] gives converged *metrics* (σ and d_W change by less than 0.5% against [70, 15, 15] wherever the class is quotable) but not a converged *model*: the feature difference is $d \geq 0.5$ in 16 of 24 A1 scenarios, and [70, 15, 15] itself differs from [90, 20, 20] by $d = 0.25$ at p_0 , dominated by the vertical-mode truncation M . Within-model (twin) conclusions are unaffected because data and inverse share the EMM; inverting *real* data would carry this bias. **V1 (4 October)** quantified it: against the converged [130, 20, 20], [50, 10, 10] gives $d_{\text{sys}} = 0.77$ to 3.22. The twin studies keep [50, 10, 10] (frozen); real-data inversion uses [130, 20, 20].

Trap. No new estimator, no SSI revisits, no EMM reformulation, no multi-board firmware work. Anything that changes the chain above invalidates G3 and all comparisons against it.

1.2 Dropped to future work

- Multi-board synchronisation experiment, `estimateClockMap.m`, `resampleImuRecord.m`, three-cornered hat. The thesis reports the derived requirements (0.37 ms offset, 0.11 ppm aligned-at-start or 8 ppm with mean offset removed, 0.42% relative gain) and a proposed architecture. The ICM-42688-P’s external clock input (31–50 kHz) and high-resolution FIFO (18-bit accelerometer, 19-bit gyro) make it the natural sensor for that architecture, even though it is not the lowest-noise part in the present measurements.
- Tank experiment. Replaced by the scenario sweeps below; a wave probe (Scenario W) remains a future-work recommendation.

1.3 Kept from the IMU work (small)

- **Shell-rate dependence check** (15 minutes): per-sample standard deviation of Tuesday’s 10/20/50 Hz rate-sweep records. Growth as $\sqrt{f_s}$ would mean the output bandwidth tracks the stream rate; a constant value means it does not, which is *consistent with* (not proof of) aliasing, since a fixed-bandwidth process has the same marginal variance at any polling rate. The 20 and 50 Hz records are lossy, so report the result as suggestive only. The aliasing question is settled by the firmware’s actual ODR and filter configuration, not by this check.
- **Measured noise** enters the programme through the SNR axis (Study R1), with sensor-specific basin-band values only.
- **Calibration and co-located transmissibility** (Study H1, reduced scope, one board): the only remaining real-hardware experiment, kept because it turns the 0.42% gain and 0.37 ms timing tolerances from requirements into demonstrated (or refuted) achievable figures.

2 Study architecture

The thesis results chapter is organised by the *kind* of question, not by script:

Category	Question	Studies
Design	How should the array be built and run?	D1 sensor count and layout; D2 record duration; D3 direction-robust layout
Robustness	Does the method tolerate realistic departures in noise, sea and prior knowledge?	R1 SNR and measured noise; R2 spectral shape; R3 noise-PSD mis-specification; R4 γ error
Domain	Which floes are identifiable at all?	A1 floe-property sweep
Model validity	What happens when a core assumption breaks, and can it be detected?	V1 forward-model mismatch; V2 directional forcing
Verification	Does the analytic route still hold away from p_0 ?	M1 Monte Carlo and nonlinear spot checks
Hardware anchor	Can real sensors meet the derived gain and timing tolerances?	H1 calibration and co-located transmissibility

2.1 Priority tiers

- **MUST** E0 inventory, E1 analytic engine, D1, A1, R1, R2, M1 (reduced).
- **SHOULD** D2, R3, R4 (cheap by-products of E1, run early as quick wins), V1, H1; V2 and D3 (added in rev. 4: D1 showed that the optimal layout depends on the wave direction, and D3 needs V2's forcing model).
- **STRETCH** extended M1, $n_s = 5-6$ exhaustive search, on-board IMU fusion.

3 Common definitions

3.1 Metric set (identical in every study)

Every scenario returns one struct from `evaluateScenario.m` with:

Metric	Definition	Source
$\sigma_{\ln \beta}, \sigma_{\ln R}$	$\sqrt{(F^{-1})_{ii}}, F = \sum_k J_k^T \Sigma_k^{-1} J_k$ (one record)	analytic
$\rho_{\beta R}, \kappa(F)$	parameter correlation; condition number of F	analytic
$\delta\theta_W, d_W$	Welch-expected parameter bias and $d = \sqrt{\delta\theta^T F \delta\theta}$	analytic
$\delta\theta_{\text{sys}}, d_{\text{sys}}$	bias from a deliberate systematic error (mis-specified S_{nn} , γ , model, direction)	analytic
Relative errors	$(\hat{\beta} - \beta)/\beta, (\hat{R} - R)/R$	M1 only
emp/pred, coverage	empirical scatter over predicted σ ; 68/95% coverage of $q \sim \chi_2^2$	M1 only
χ^2/dof , iterations	nonlinear fit quality and convergence	M1 only
Invalid-bin fraction	bins where $\hat{S}_{rr} - S_{nn}$ falls below the rejection threshold (analytic proxy: expected S_{rr}/S_{nn})	both
Minimum coherence	$\min_{j,k} \gamma_{jr}^2(\omega_k)$ in the band	analytic

3.2 Pre-registered classification

Fixed now, before any sweep is run, for one record of the reference duration (109 min) unless a study says otherwise:

With

$$s_{\max} = \max(\sigma_{\ln \beta}, \sigma_{\ln R}), \quad \delta\theta_{\text{tot}} = \delta\theta_W + \delta\theta_{\text{sys}}, \quad d = \sqrt{\delta\theta_{\text{tot}}^T F \delta\theta_{\text{tot}}},$$

the classes are exhaustive and applied in order:

Class	Condition
Identifiable	$s_{\max} \leq 1\%$, $d \leq 0.5$ and $\kappa(F) \leq 10^8$
Marginal	not Identifiable, and $s_{\max} \leq 5\%$, $d \leq 1$ and $\kappa(F) \leq 10^8$
Not identifiable	otherwise

Amendment (2 October, A1). A scenario whose forward model fails its validity checks (node grid, truncation metric convergence, α window, or a singular F at either truncation) is labelled *Model check failed*, never *Not identifiable*; where the class is the same at both truncations it is reported as robust with σ not quoted. The bias vectors are added, not their distances maximised, because systematic errors can add or partly cancel. $\delta\theta_{\text{sys}} = 0$ in studies without a deliberate systematic error.

For systematic-error studies (R3, R4, V1, V2) the tolerance is the error size at which $|d_{\text{sys}}|$ first reaches 0.5 and 1, and the induced $|\delta\beta/\beta|$ first reaches 1% (same first-crossing rule as the Monday tolerance study). Thresholds may be reported at other levels later, but these are the headline ones.

3.3 Noise levels used throughout

One-sided, per rad/s, vertical, basin band, from the 16.9 h record (MIZ-band z values are *not* used: they contain a common component):

Case	$S_{nn,\omega}, (\text{m s}^{-2})^2/(\text{rad/s})$	σ_n at $\Delta t = 0.1 \text{ s}, \text{m s}^{-2}$
Datasheet grade ($65 \mu\text{g}/\sqrt{\text{Hz}}$)	6.5×10^{-8}	1.4×10^{-3}
Three on-board IMUs fused (stretch)	$\approx 5.0 \times 10^{-7}$	4.0×10^{-3}
LSM6DSV16X	8.0×10^{-7}	5.0×10^{-3}
G3 twin assumption (10% stress)	1.416×10^{-6}	6.7×10^{-3}
ICM-42688-P	1.52×10^{-6}	6.9×10^{-3}
LSM6DS3TR-C	1.24×10^{-5}	2.0×10^{-2}

$\sigma_n^2 = \pi S_{nn,\omega} / \Delta t$. The datasheet-grade row uses $65 \mu\text{g}/\sqrt{\text{Hz}}$, the middle of the quoted 60–70 range (Section 3.4). The fused value assumes independent noise between the three co-located sensors, which the basin-band coherence (at chance level) supports. Arrays are homogeneous (one sensor type throughout) unless a study states otherwise; each channel carries its own S_{nn} in Σ_k .

3.4 Datasheet reference values

Only the specifications that bear on noise, calibration, timing and sensor choice; for the thesis sensor table and for H1.

Property	LSM6DS3TR-C	LSM6DSV16X	ICM-42688-P
Accel noise density	$90 \mu\text{g}/\sqrt{\text{Hz}}$ ($\pm 2/4/8$ g)	60 (high-performance), 100 (normal)	65 (x, y), 70 (z)
Gyro noise density	$5 \text{ mdps}/\sqrt{\text{Hz}}$	2.8	2.8
Max ODR (accel, gyro)	6.664 kHz	7.68 kHz	32 kHz
FIFO	4 kB, timestamp storage	4.5 kB, tagging and timestamps	2 kB; high-resolution mode 18-bit accel, 19-bit gyro
Timing features	data-ready, DEN/sync	data-ready, sync, timestamps	external clock input 31–50 kHz , FSYNC, timestamps
Zero- g offset (typ.)	± 40 mg	± 12 mg	± 20 mg
Zero- g temperature drift	$\pm 0.5 \text{ mg}/^\circ\text{C}$	$\pm 0.07 \text{ mg}/^\circ\text{C}$	$\pm 0.15 \text{ mg}/^\circ\text{C}$
Temperature sensor	yes	yes	yes
Measured z basin ASD (16.9 h)	900	229	315

- **Convention.** The datasheets do not state a one- or two-sided convention. Their densities are used as a reference ASD specification; the measured values use the one-sided convention of this project. The measured/datasheet ratios (about 3–10) are board-level results under an unconfirmed ODR, filter and acquisition mode, not sensor underperformance.
- **Resolution.** All three parts output at least 16-bit data; the present gyro limitation is the shell’s 0.001 rad/s print step, a firmware/logging limitation, not an IMU one.
- **Zero- g offsets** of ± 12 to ± 40 mg (up to 0.39 m s^{-2}) alone could explain the observed $|a|$ deviations, which is why H1a separates offset, scale and non-orthogonality.
- **Drift consistency.** The overnight first-to-last-hour z changes imply, at the typical coefficients, about 4°C for the LSM6DS3TR-C (2.2 mg) but about 14°C for the LSM6DSV16X (1 mg), implausible indoors. The LSM6DSV16X drift is therefore probably not offset temperature drift alone. Record as an observation; it strengthens the case for logging temperature.

4 E0: inventory and results register **MUST**

Goal. Make every completed study citable in the thesis without re-deriving it. Half a day, hard time-box.

Create `Docs/resultsRegister.md` (or a LaTeX table) with one row per completed study:

ID	Study	Headline result	Status
C1	Forward model (EMM) validation		thesis-grade / rerun
C2	SSI validation ladder, frequency proximity, amplitude imbalance		
C3	Sensor placement, Gramian criteria, Level 2C exhaustive check (40 920 layouts)	$\lambda_{\min}$ best in that experiment (cross-MAC 0.033 vs 0.216)	
C4	EMM identifiability, three-parameter Jacobian (γ weak direction)		
C5	Original EMM inverse twin		
C6	SSI-to-transmissibility pivot; FRF reconnaissance (no peak > 0.3 dB)		
C7	Output-only information bound (5c/5d)	$\lambda = [0.834, 0.097]; 1.72\times, 2.98\times$	
C8	G2 finite-record estimator	bias is deterministic spectral-window effect	
C9	U1 covariance treatments; $L = 2048$ refinement	$d_{\text{pred}} 3.68 \rightarrow 0.91 \rightarrow 0.26$	
C10	G3 nonlinear inversion and jack-knife	$\sigma = 0.46\%, 0.088\%$; coverage 0.93	
C11	Measurement-tolerance study	0.37 ms, 0.11/8 ppm, 0.42%	
C12	Mission-physics regime screening		
C13	Shell-rate sweep, PSD/Allan tools, 16.9 h IMU record	basin S_{nn} table	

Columns in the real register: script(s), results file, headline numbers, figure file, thesis section, status (*thesis-grade*, *figure needs regenerating*, *number needs checking*). Mark every figure that needs regenerating now; do not regenerate during E0.

5 E1: analytic Fisher and bias engine **MUST**

Goal. One validated function that turns a scenario (floe, layout, reference, sea state, noise, duration) into the metric set without Monte Carlo. Every later study is a loop over this engine.

5.1 Why the analytic route is legitimate

G3 established at p_0 and at two displaced truths that (i) the linearised Fisher prediction matches empirical scatter to about 10%, (ii) the Welch-expected bias matches the Monte Carlo and validation mean offsets within their standard errors, and (iii) the nonlinear solver lands within 0.003σ of the linearised estimate. The programme therefore uses analytic quantities for sweeps and reserves Monte Carlo for spot checks (M1). That is a defensible choice only if the engine reproduces G3 first. *Rev. 6:* M1 has since confirmed the analytic predictions in four stress cases away from p_0 (Section 12).

5.2 Components

1. **Field sampler.** EMM responses $H(\omega; \theta, \mathbf{x})$ at arbitrary points $\mathbf{x}$ (the 33-point candidate grid and rotated copies for V2), on the 0.2 rad/s node grid, spline-interpolated to the retained bins. This grid was validated only at p_0 and truths A and B (max relative error 1.7×10^{-5} , $d_{\text{interp}} \approx 0$); records were generated on 0.1 rad/s. Away from p_0 the 0.2 grid is a computational approximation: check it against 0.1 rad/s for every A1 scenario (require $d_{\text{interp}} < 0.1$) and fall back to 0.1 where it fails. Cache by (θ, γ) .

2. **Jacobian.** Central differences in $(\ln \beta, \ln R)$ and, for R4, $\ln \gamma$ ($h = 10^{-3}$, as U1). One set of solves serves every layout and reference choice. **Sensor positions are fixed physical coordinates during differentiation:** when R is perturbed, the sensors do not move with the trial floe (a layout defined as $r_i = \xi_i R_{\text{true}}$ is converted to coordinates once, at the truth). Otherwise $\partial T / \partial \ln R$ differentiates the layout as well as the floe. This matches the existing R -sensitivity work.
3. **Analytic transmissibility covariance.** Full real covariance $\Sigma_k \in \mathbb{R}^{2(n_s-1) \times 2(n_s-1)}$ per bin, including Re/Im cross terms and the correlation between transmissibilities that share the reference. Derivation: first-order (delta-method) propagation of the complex-Wishart covariance of the Welch spectral matrix $\hat{\mathbf{S}}$, with the overlap-corrected number of averages, through

$$\delta T_j = \frac{\delta S_{jr} - T_j \delta S_{rr}}{S_{rr} - S_{nn}},$$
 written in the real parametrisation $[\Re \delta T; \Im \delta T]$ directly, because δS_{rr} is real and the error is not circular. Sensor noise enters through $\mathbf{S} = \mathbf{H}\mathbf{H}^H S_u + \text{diag}(S_{nn,j})$. **Two refinements found while building it (1 October):** (i) the spectral matrix must be the *window-smoothed* one, $\hat{\mathbf{S}}_k = \sum_q K_L(\omega_k - \omega_q) G_q \mathbf{H}_q \mathbf{H}_q^H$ (the kernel of the Welch-bias formula); at short L it is no longer rank one, coherence drops and the variance rises, which the point form misses (it was 12% optimistic in σ at $L = 512$); (ii) $N_{\text{eff}} - 1$ degrees of freedom, a finite-sample correction validated against direct complex-Gaussian simulation. With both, σ matches the cached Monte Carlo within 2% at $L = 512, 1024$ and 2048 , and every covariance element agrees to within its Monte Carlo sampling error (standardised errors: rms 1.02). Rejected bins are removed from F ; γ derivatives are nuisance sensitivities, never estimated; a singular F is reported as Not identifiable. The d_W acceptance tolerance (10%) is a logged amendment to the pre-registration, added after the first comparison (analytic d_W 6–7% below the Monte Carlo targets).
4. **Bias projector.** Welch-expected feature bias b_W (Saturday’s window formula, already in U1 code), and generic $\delta \theta = F^{-1} J^T \Sigma^{-1} b$, $d = \sqrt{\delta \theta^T F \delta \theta}$ for any systematic feature error b .
5. **Scenario evaluator.** `evaluateScenario(scen)` returning the metric struct of Section 3.1.

5.3 Validation gate (must pass before any study)

1. Σ_k analytic versus U1’s 200-record Monte Carlo Σ_k at p_0 , setting F: element-wise agreement within Monte Carlo error (about 10% on variances), and correlation structure reproduced.
2. $\sigma(\ln \beta, \ln R)$ from the analytic engine reproduces F: 0.00462, 0.00088 within 5%.
3. d_W reproduces 0.26 (F), 0.91 (B) and 3.68 (A at $L = 512$).
4. At truths A and B: oracle σ 0.00498/0.00095 and 0.00431/0.00082 within 5%.
5. Invariance checks: permuting non-reference sensors leaves F unchanged; doubling N halves F^{-1} .

Trap. Do not approximate Σ_T by independent scalar variances per transmissibility. Shared-reference correlation is large and ignoring it makes the layout search optimistic. Keep the decorrelated every-second-bin grid: using every bin as independent was shown in U1 (treatment D) to create fictitious information. If the hand-derived Σ_k fails gate 1 by more than 20% and cannot be fixed within the day, keep the same delta method but compute the Jacobian of the map from the real and imaginary spectral-matrix entries to $[\Re T; \Im T]$ *numerically* (finite differences on the explicit ratio formula), and combine it with the known complex-Wishart covariance of $\hat{\mathbf{S}}$. This preserves each layout’s own correlation structure. Do *not* scale the p_0 Monte Carlo Σ_k by an SNR ratio: changing the sensors or the reference changes the correlation structure, not only its magnitude, so that shortcut is invalid for D1.

6 Quick wins from the engine **SHOULD**

These cost minutes once E1 passes, and produce directly usable experimental requirements.

6.1 R4: uncertainty in the known γ

The inverse fixes $\gamma_{\text{inv}} = \gamma_{\text{true}}(1 + \epsilon_\gamma)$; the solver compensates through β and R . Linear prediction

$$\delta\theta_\gamma = F^{-1} J^T \Sigma^{-1} \frac{\partial f_T}{\partial \ln \gamma} \epsilon_\gamma,$$

for $\epsilon_\gamma = \pm 0.5, \pm 1, \pm 2, \pm 5, \pm 10, \pm 20\%$, then a nonlinear inversion of the noiseless Welch expectation at $\pm 5\%$ and $\pm 20\%$ to confirm linearity. Earlier FRF-based estimate for comparison: 0.057 (β) and 0.008 (R) per unit $\ln \gamma$ error; recompute with the transmissibility features.

Deliverable: required draught (thickness) accuracy for $|\delta\beta/\beta| \leq 1\%$ and for $d_{\text{sys}} \leq 0.5$. Report alongside the $\beta \propto h^3$ ground-truth caveat: 1% in h is 3% in the independent β reference.

Result (1 October). γ to $\pm 1.64\%$ for $d_{\text{sys}} \leq 0.5$ and $\pm 10.6\%$ for $|\delta\beta/\beta| \leq 1\%$ (LSM6DSV16X): determine γ by weighing. `SHARC_R4_Gamma_Tolerance.tex`.

6.2 D2: record duration

For fixed L , $\sigma \propto N^{-1/2}$ while $\delta\theta_W$ is fixed, so $d_W \propto \sqrt{N}$. Evaluate $T_{\text{rec}} = 10, 20, 30, 60, 109, 180, 240$ min for each sensor type at $L = 512, 1024, 2048, 4096$. A combination is admitted only if it has at least $K \geq 8$ Welch segments ($K = \lfloor 2N/L \rfloor - 1$ at 50% overlap): at 10 min and $f_s \approx 10$ Hz, $L = 2048$ gives only about 4 segments and $L = 4096$ about 1, which are not meaningful covariance cases. Report, for the admitted combinations,

$$\text{RMSE} = \sqrt{\sigma^2 + \delta\theta_W^2}.$$

Deliverables: (i) minimum record length for $\sigma_{\ln \beta} \leq 1\%$ per sensor type; (ii) the L that minimises RMSE at each duration, which replaces the local rule $d \approx 0.26\sqrt{N}/65536 (2048/L)^2$ with a checked design curve; (iii) the duration at which $L = 2048$ stops being adequate ($d_W > 0.5$); (iv) the finest spectral resolution that is feasible at each duration under the $K \geq 8$ rule.

Result (2 October). $\sigma_{\ln \beta} \leq 1\%$ needs 17.1 min at $L = 2048$ (LSM6DSV16X); the Identifiable window is 17.1 to 281.5 min; $L = 512$ is never Identifiable; the datasheet-grade case is only Marginal at 109 min because d_W grows as $\sqrt{N}$. `SHARC_D2_Record_Duration.tex`.

6.3 R3: noise-PSD mis-specification

The estimator subtracts an assumed S_{nn} . With $S_{nn,\text{assumed}} = (1 + \epsilon)S_{nn,\text{true}}$, the denominator error is deterministic: $b_j \approx T_j \epsilon S_{nn} / (S_{rr} - S_{nn})$. Sweep $\epsilon = \pm 10, \pm 30, \pm 50, -100\%$ ($-100\% =$ no correction) at each measured noise level, plus one coloured case using the measured low-frequency shape of the LSM6DS3TR-C spectrum. **Deliverable:** how accurately the noise floor must be known, per sensor type.

Result (2 October). (bias computed with the Welch-smoothed T_W) the floor must be known to $\pm 30.6\%$ for $d_{\text{sys}} \leq 0.5$ (LSM6DSV16X), $\pm 7.7\%$ for the LSM6DS3TR-C; no correction gives $d_{\text{sys}} = 1.63$, so the correction is required; the measured noise colour is negligible ($d_{\text{sys}} \leq 0.06$). `SHARC_R3_Noise_Floor.tex`.

7 D1: sensor count and layout **MUST**

Goal. Answer “how much does each additional sensor buy?” with the metric that the inverse actually uses, and connect it to the earlier Gramian work.

7.1 Design

- **Candidate set:** the same 33-point grid as Level 2C, nominal floe p_0 , setting F, LSM6DSV16X noise (then repeated at ICM and G3 levels).

- **Admissibility (fixed before the search):** radius $\leq 0.9R_0$ (edge clearance, as Level 2C) and within one buoy radius of the edge; minimum centre-to-centre spacing equal to the buoy diameter, 0.10 m (provisional: the SHARC buoy is not final; 0.05 and 0.15 m reported alongside). At p_0 ($R_{\text{phys}} = 0.72$ m) the closest grid points are 0.144 m apart, so the constraint does not bind: all 40 920 four-sensor layouts are admissible. It binds for the 0.2 m floes of A1. Report the unconstrained optimum as well, so the cost of the constraints is visible.
- **Exhaustive search** over admissible layouts for $n_s = 2, 3, 4$ ($\binom{33}{2} = 528$, $\binom{33}{3} = 5456$, $\binom{33}{4} = 40\,920$ before constraints), with the reference optimised within each layout (n_s choices). $n_s = 5$ (237 336 layouts) is stretch; use it if vectorised F evaluation keeps it under an hour, otherwise greedy plus pairwise exchange from the $n_s = 4$ optimum.
- $n_s = 1$ is reported only through the Gramian (a single sensor has no transmissibility).

7.2 Criteria

Primary: **A-optimal**, minimise $\text{tr}(F^{-1}) = \sigma_{\ln \beta}^2 + \sigma_{\ln R}^2$; it is dominated by the weak parameter β , which is the science target. Also reported: D-optimal ($\log \det F$) and E-optimal ($\lambda_{\min}(F)$). These are the parameter-space counterparts of the trace, log-determinant and $\lambda_{\min}$ criteria compared on the Gramian in Level 2C, so the earlier cost-function study carries over as the criterion comparison here.

7.3 Gramian comparison

For each n_s , take the Level 2C Gramian-optimal layout ($\lambda_{\min}$ criterion) and compute its Fisher efficiency

$$\eta = \frac{\text{tr}(F_{\text{Fisher opt}}^{-1})}{\text{tr}(F_{\text{Gramian opt}}^{-1})} \in (0, 1],$$

and the rank of the Gramian-optimal layout in the Fisher ordering. Either outcome is a result: agreement justifies the cheap Gramian as a design proxy; disagreement shows that modal observability is not parameter identifiability for this inverse.

7.4 Outputs

- Best $\sigma_{\ln \beta}$, $\sigma_{\ln R}$, d_W and $\kappa(F)$ against n_s , with the chosen layouts drawn on the floe.
- Distribution of $\sigma_{\ln \beta}$ over all admissible layouts per n_s (how forgiving is placement?).
- Best reference position per n_s ; confirm or overturn s26.
- Level 2C (s2, s6, s14, s26) placed in the $n_s = 4$ ranking.

Pre-registered success criterion. an additional sensor is “worth it” if it lowers $\sigma_{\ln \beta}$ by at least 20% at the optimum. The recommended count is the smallest n_s that reaches the Identifiable class at LSM6DSV16X noise.

Result (2 October). $n_s = 2$ **recommended:** s6 with reference s18 on the incidence axis, $\sigma_{\ln \beta} = 0.46\%$, Identifiable at all three noise levels. Gains 19.6% ($n_s = 3$) and 10.2% ($n_s = 4$), below the 20% rule; optima nested (s6, s18, s14, s26). The reference choice changes $\text{tr}(F^{-1})$ by a median factor of 1.00; s26 confirmed for Level 2C, which ranks 167 of 40 920 ($\eta = 0.89$). Gramian $\lambda_{\min}$ designs: $\eta = 0.21$ to 0.35, so modal observability is not parameter identifiability here. Extra sensors buy placement robustness: layouts within 10% of the optimum number 4, 13 and 101 for $n_s = 2, 3, 4$. **Caveat that motivates D3:** every optimum lies on the axis of a single, known wave direction. SHARC_D1_Sensor_Layout.tex.

8 A1: floe-property and domain sweep **MUST**

Goal. Replace “it works at p_0 and two neighbours” with a map of which floe classes are identifiable.

8.1 Physical parametrisation

Scenarios are specified physically and converted with the EMM’s own non-dimensionalisation:

$$D = \frac{Eh^3}{12(1-\nu^2)}, \quad \beta = \frac{D}{\rho_w g H^4}, \quad \gamma = \frac{\rho_i h}{\rho_w H}, \quad R = \frac{R_{\text{phys}}}{H}.$$

Each scenario carries its own known γ ; the inverse estimates only $[\ln \beta, \ln R]$. Each physical scenario uses *one* water depth H consistently in β , γ , R , the frequency scaling and the dispersion relation. Sensor positions are placed once per scenario (for example $r_i = \xi_i R_{\text{true}}$ on the Level 2C fractions) and then held fixed as physical coordinates for the Jacobian and the inversion (Section E1).

8.2 Scenario set (20–25, fixed before running)

1. **Parameter-space ring** (8): $\beta/\beta_0 \in \{0.25, 0.5, 2, 4\}$ at R_0 , and $R/R_0 \in \{0.5, 0.75, 1.5, 2\}$ at β_0 . Isolates each direction.
2. **Corners** (4): $(\beta/\beta_0, R/R_0) \in \{0.5, 2\}^2$.
3. **Physical classes** (12), fixed in `defineStudyScenarios.m`: the basin reference; polypropylene discs (SG 0.91, $E \approx 1.5$ GPa, $\nu = 0.4$): 3 mm at the p_0 radius (a realisable p_0), 5 and 10 mm at 40 cm diameter (the model floe’s core sheets), 5 mm at 1 m diameter; the SHARC model floe (44 cm diameter, 6 cm maximum thickness, elliptical, SG 0.91, stiffness unknown) as a volume-equivalent 4 cm plate at $E_{\text{eff}} = 0.5, 1.5, 3$ GPa, *provisional* until the CAD model and weighing; and four MIZ floes ($R = 10, 25, 50$ m at $h = 1$ m, $E = 6$ GPa; $R = 25$ m at $h = 0.5$ m, $E = 3$ GPa).
4. **Hypothesis recorded before running** (not a result until A1 computes F): the model floe has $R/\ell_f \approx 0.19\text{--}0.29$ against 4.6 at p_0 (β is 550–3300 times β_0): it is close to a rigid body in the basin band, so we expect its β sensitivity to be weak, while R may remain identifiable. Its thickness-to-diameter ratio also makes thin-plate theory marginal.

8.3 The frequency band per scenario

The EMM’s length scale is the water depth ($\alpha = H\omega^2/g$, $R = R_{\text{phys}}/H$), and its truncation is validated for $\alpha \in [1.7, 13.85]$; the basin band 3.178–8.301 rad/s is $\alpha \in [1.94, 13.2]$. Fixed at pre-registration (`defineStudyScenarios.m`):

- **Finite depth (basin and tank floes):** the basin band itself, i.e. the same validated α window. A1 therefore answers *which floes are identifiable under the fixed, validated basin excitation band*, an engineering question; it does not isolate floe properties, since changing R also changes kR . The earlier floe-relative rule (same kR range as p_0) was dropped: for the 0.2 m floes it needs ω up to about 15 rad/s, $\alpha \approx 40$, far outside the validated range.
- **Deep water (MIZ classes):** periods 6–12.5 s, with the effective depth set at 12.5 s by `screenFloeRegime.m` so that $\alpha \leq 13.85$.

For every scenario: rerun the EMM truncation convergence check (nominal versus a higher truncation) and the 0.2 versus 0.1 rad/s node-grid check, and exclude or flag scenarios whose band leaves the validated EMM range. MIZ scenarios are labelled *regime and domain screening* unless their depth, frequency and truncation combination passes these checks.

Trap. Without a per-scenario band, the sweep measures whether the basin band happens to suit a floe, not whether the floe is identifiable. Without the truncation check, a failure may be the forward model, not the inverse.

8.4 Cost

Per scenario about $27 \text{ nodes} \times (1 + 4 \text{ for the } \theta \text{ Jacobian} + 2 \text{ for } \gamma) \approx 190$ EMM solves, plus a truncation check. G3 timings (13 model evaluations of about 27 solves in 349 s) suggest about 1 s per solve, so roughly 3–6 minutes per scenario and 2–3 hours for the set. Run the set as one batch.

8.5 Outputs

Identifiability map in $(\beta/\beta_0, R/R_0)$ and in physical terms (h, E, R_{phys}) , with the class of each scenario; σ , d_W , $\kappa(F)$ and $\rho_{\beta R}$ versus kR and R/ℓ_f (the non-dimensional groups that should organise the result).

Logged amendments before the batch (2 October): the truncation check is a *metric* gate (σ within 1%, d_W within 0.02, same class), replacing a feature-bias gate after the p_0 pre-flight; the feature bias is reported for V1 but not gated. The MIZ floes are evaluated Froude-scaled to the basin (the EMM sees only α, β, γ, R), with the 6–12.5 s band cut to the basin band. Poisson’s ratio is passed to the EMM per scenario.

Result (3 October). 12 Identifiable, 6 Marginal, 6 Not identifiable (class robust to the truncation, σ not converged). **Lower boundary, physical:** below $R/\ell_f \approx 1$ the stiffness is lost ($\sigma_{\ln \beta}$ 14 to 440%) while R stays measurable; the cliff lies between $R/\ell_f = 0.96$ and 1.64. **Upper boundary, estimator-limited:** at $kR \approx 3.7$, $\sigma_{\ln \beta} \approx 0.2\%$ but $d_W \approx 0.6$. In between, kR sets the precision; β/β_0 from 0.25 to 4 moves $\sigma_{\ln \beta}$ only from 0.35 to 0.42%. The model-floe hypothesis is confirmed (rigid in the band); a basin stiffness test needs a thin plate of about 1 m diameter. MIZ (scaled): 10 m Not identifiable, 25 m on the boundary, 50 m Identifiable. Truncation: see Section 1.1. An EMM damped-root fallback was added to `dispersionRoots.m` for one scenario (original roots unchanged). `SHARC_A1_Floe_Domain.tex`.

9 R1 and R2: SNR, measured noise and spectral shape **MUST**

9.1 R1: SNR axis with measured noise

The system is linear, so only the ratio S_{nn}/S_{signal} matters, and $S_{\text{signal}} \propto H_s^2$. Halving the noise *PSD* is equivalent to increasing the wave amplitude by $\sqrt{2}$; halving the noise *ASD* is equivalent to doubling it. Noise level and sea-state *amplitude* are therefore one axis, not two studies. Sweep $S_{nn,\omega}$ continuously (logarithmic, 10^{-8} to 10^{-4}) at fixed JONSWAP ($H_s = 0.05$ m, $\omega_p = 5$ rad/s) and mark the measured points of Section 3.3.

Deliverables: $\sigma_{\ln \beta}$, $\sigma_{\ln R}$, invalid-bin fraction and class versus S_{nn} ; the noise level at which each class boundary is crossed; the ratio between the G3 assumption and each real sensor. This is the analytic version of the planned measured-noise twin rerun, which M1 then confirms for LSM6DSV16X.

Result (3 October). an Identifiable **window**, $1.868 \times 10^{-7} \leq S_{nn} \leq 6.547 \times 10^{-6}$ (110.4 to 653.8 $\mu\text{g}/\sqrt{\text{Hz}}$; for the LSM6DSV16X, $0.018 \leq H_s \leq 0.104$ m at this shape). Below it the inverse is bias-limited ($d_W > 0.5$), above it precision-limited ($\sigma_{\ln \beta} > 1\%$); the quiet end is an estimator limit at $L = 2048$, mitigated by a finer resolution, not by more noise. Inside: fused, LSM6DSV16X, G3, ICM-42688-P (PSD/G3 = 1.07, so the G3 assumption was realistic); outside: datasheet grade (bias) and LSM6DS3TR-C (precision). 0% invalid bins over the sweep. `SHARC_R1_SNR.tex`.

9.2 R2: spectral shape at fixed variance

Shape moves information between bins and changes the Welch bias. At fixed H_s :

- peak position $\omega_p \in \{3.5, 4.25, 5, 6, 7.5\}$ rad/s across the informative band (most of U1’s Fisher information came from 4.7–6.6 rad/s);
- peakedness $\gamma_J \in \{1, 3.3, 7\}$;
- one bimodal (swell plus wind sea) case.

Deliverables: σ and d_W per shape; whether $L = 2048$ remains adequate for narrow and bimodal seas (Sunday flagged these as the cases that steepen the input weighting); the demonstration that T_j is spectrum-independent in theory while $\hat{T}_j$ is not in practice.

Clarification (logged before the run): “fixed H_s ” is implemented as fixed *in-band variance*, so R2 changes where the usable energy lies and R1 covers how much there is; each sea is run at $L = 1024, 2048, 4096$.

Result (3 October). the exact T and J are identical for all 16 seas, yet at $L = 2048$ $\sigma_{\ln \beta}$ ranges from 0.161 to 0.766% (factor 4.8 at equal in-band energy) and d_W from 0.048 to 0.359. **All 16 Identifiable at $L = 2048$; $L = 1024$ only 3 of 16; $L = 4096$ gives the lowest RMSE for all 16.** Energy in the upper band is most informative; a narrow sea at the lower band edge is worst; the bimodal sea is benign. `SHARC_R2_Spectral_Shape.tex`.

10 V1 and V2: model validity **SHOULD**

10.1 V1: forward-model mismatch (breaking the inverse crime)

Every recovery study so far generates and inverts with the same EMM. V1 generates the noiseless Welch expectation with a perturbed forward model and inverts with the nominal one:

- higher EMM truncation for the data. **Revised (rev. 5, from A1):** $[70, 15, 15]$ is not a valid truth ($d = 0.25$ to $[90, 20, 20]$ at p_0). Before V1 runs, extend the ladder in M (which dominates) until one step changes the features by $d < 0.1$, and set `cfg.V1.truncationTruth` to that level as a logged change. Report the bias of $[50, 10, 10]$ against it at p_0 and at the large- kR A1 floes, where A1 found it largest;
- water depth mis-specified by $\pm 2\%$ and $\pm 5\%$;
- sensor position errors of ± 1 and ± 2 cm (radial), which the field placement will have;
- γ mismatch is already covered by R4.

Report d_{sys} and $\delta\beta/\beta$, $\delta R/R$. **Point of the study:** separate estimator/inversion error (G3) from forward-model discrepancy, which the thesis must distinguish.

Clarifications logged before the run (4 October; no threshold changed). $b_{\text{sys}} = f_W(\text{truth}) - f_W(\text{nominal})$, both Welch expectations from one function, projected with the nominal J , Σ , F ; the class uses d_{tot} from the vector sum $\delta\theta_W + \delta\theta_{\text{sys}}$. Stage 0 ladder $M = 90, 110, \dots, 210$ with $[P\ N] = [20\ 20]$ at p_0 , plus the last step at the large- kR floes and a $P = 30$ check. Depth is physically consistent ($\beta \propto H^{-4}$; $\gamma, R, r_i \propto H^{-1}$) with the physical sea and band fixed; the α range is reported per case. The radial error is a common signed shift of every sensor. Tolerances are first crossings per side of $d_{\text{sys}} = 0.5, 1$ and $|\delta\beta/\beta| = 1\%$; *pre-run correction*: bisection in $\log_{10} |\text{error}|$ to 0.005 decade (an absolute tolerance could not resolve crossings near 0.05 mm). Nonlinear diagnostic on the largest- d_{sys} case.

Result (4 October). Stage 0: $d = 0.125$ (M 90 to 110), 0.079 (110 to 130), so truth $[130, 20, 20]$; converged also at the large- kR floes ($d \leq 0.069$) and in P ($d = 0.050$). **Truncation:** $d_{\text{sys}} = 1.02$ (p_0 , Identifiable to Marginal), 1.83 and 3.22 (two large- kR floes, Marginal to Not identifiable), 0.77 (unchanged) for shifts below 0.4% in β : converged metrics, not a converged model. **Depth $\pm 5\%$:** $d_{\text{sys}} \leq 0.205$, all Identifiable; the tolerance exceeds $\pm 5\%$ (retained α inside the validated window, support extrapolated). **Radial position:** $d_{\text{sys}} = 0.5$ at 0.050 mm, 1 at 0.10 mm, $|\delta\beta/\beta| = 1\%$ at 0.29 mm, symmetric in sign; the ± 1 and ± 2 cm cases give $d_{\text{sys}} \approx 100$ to 200 and are far outside the linear regime (nonlinear at -2 cm: $+80.5\%$ against linear $+107\%$). Biases add as vectors (at p_0 they partly cancel: $d_{\text{tot}} = 0.905$). **Requirements:** real data inverted with $[130, 20, 20]$; depth to $\pm 5\%$ suffices; sensor radial coordinates relative to the model geometry known or jointly estimated to sub-millimetre precision (calibration, photogrammetry or position parameters in the inverse). Run `V1_20261004_121316`, commit 1711fb3, 59 min. `SHARC_V1_Model_Mismatch.tex`.

10.2 V2: directional and multiple-source forcing

With two independent incident components, $\mathbf{y} = \mathbf{H}_1 u_1 + \mathbf{H}_2 u_2$, the spectral matrix is rank two and S_{jr}/S_{rr} is no longer H_j/H_r . For a circular floe, a second direction ϕ is the nominal solution evaluated at sensor positions rotated by $-\phi$, so the engine's field sampler supplies it. Sweep $\phi \in \{15^\circ, 30^\circ, 60^\circ, 90^\circ\}$ and energy fraction of the second component $\in \{5, 10, 25, 50\%\}$. Ask two questions:

1. **Bias:** d_{sys} and $\delta\beta/\beta$ from the Welch expectation.
2. **Detectability:** the coherence drop $1 - \gamma_{jr}^2$ against the coherence estimator's own scatter at N_{eff} (and against the drop that sensor noise alone causes). If the drop becomes significant before d_{sys} reaches 0.5, coherence is a usable deployment-validity diagnostic.

The second question is the stronger thesis result; answer it even if only two angles are run.

Interface for D3 (rev. 4). Build the forcing model first, as one function returning the (window-smoothed) spectral matrix at any set of fixed physical points for a list of components (heading, energy fraction, spectrum), $\mathbf{S} = \sum_c \mathbf{H}_c \mathbf{H}_c^H S_{u,c} + \text{diag}(S_{nn,j})$, and optionally a discretised directional spread (for example $\cos^{2s}$ about a mean heading). V2 and D3 then share it, so the bias that V2 reports is exactly the bias that D3 includes in its layout ranking. V2 was promoted from STRETCH to SHOULD for this reason.

11 D3: direction-robust sensor count and layout **SHOULD**

Goal. Repeat D1 without assuming a single, known wave direction, and find out whether two sensors still suffice, which layout is robust, and how much the D1 optimum and Level 2C lose when the sea is directional. Added in rev. 4 after the D1 result.

11.1 Why D1 is not the final answer

D1 forces the floe from one known heading, so the transmissibility varies most along the incidence axis and every optimum lies on it. In the field the heading is estimated, not known, and the sea has spread or several components. Two effects follow, and D3 treats them separately:

- **Unknown heading (D3a).** The forcing is still unidirectional, so $T_j = H_j/H_r$ still holds, but the heading ψ is an extra unknown. It enters F as a nuisance parameter: with $\theta = [\ln \beta, \ln R, \psi]$, the (β, R) block of F^{-1} (Schur complement) gives the uncertainty after estimating the heading. A layout can also be good for one heading and poor for another, so its performance is evaluated over all headings.
- **Multi-directional sea (D3b).** With two or more independent components the spectral matrix is rank two or higher and $S_{jr}/S_{rr} \neq H_j/H_r$ (V2). The single-direction model is then wrong, so a layout must be judged on its bias as well as its variance: the V2 bias $\delta\theta_{\text{dir}}$ is projected with each layout's own F , added to $\delta\theta_W$, and the class uses d_{tot} . The covariance Σ_k is computed from the rank-two $\mathbf{S}$ (lower coherence, larger variance) through the same Wishart delta method.

11.2 Design

- **Candidates, setting and admissibility** as D1 (33 points, p_0 , setting F, 0.10 m spacing; LSM6DSV16X headline, ICM-42688-P and G3 repeated). Exhaustive $n_s = 2, 3, 4$, reference optimised.
- **Headings.** $\psi \in \{0^\circ, 15^\circ, \dots, 345^\circ\}$. For a circular floe a heading ψ is the nominal field at points rotated by $-\psi$; the grid has eight-fold symmetry, so the 45° multiples are pure index permutations of the D1 field (no new solves), and the 15° and 30° offsets need the field at two rotated copies of the grid (64 new points, cached; the centre is shared). $\partial T/\partial\psi$ by central differences in the rotation.
- **Sea states for D3b** (from V2, fixed in `defineStudyScenarios.m` before the run): a second component at $\phi \in \{30^\circ, 60^\circ, 90^\circ\}$ with 10 and 25% of the energy, and one $\cos^{2s}$ spread case.

11.3 Criteria

For each layout, with the best reference for each condition:

- **Average A** (primary): $\bar{A} = \frac{1}{N_\psi} \sum_\psi \text{tr}(F_\psi^{-1})_{\beta R}$, heading uniform; for D3a with the heading as nuisance, for D3b averaged over the sea states as well.
- **Worst-case A** over headings (and sea states): the guarantee for a deployment that cannot choose its orientation.
- **Bias-constrained**: only layouts with $d_{\text{tot}} \leq 0.5$ in every D3b case are ranked; report how many pass.
- **Efficiency of earlier designs**: the D1 optimum, its rotations, and Level 2C under each criterion.

Pre-registered success criterion. the same rules as D1: an additional sensor is worth it if it lowers the worst-case $\sigma_{\ln \beta}$ by at least 20% at the optimum; the recommended count is the smallest n_s whose worst-case optimum is Identifiable at LSM6DSV16X noise, with d_{tot} including the D3b bias. If no layout passes the bias constraint at some n_s , that is the result (directional seas break the single-input model regardless of layout) and D3 reports which sea states cause it.

11.4 Outputs

- Robust optimal layouts for $n_s = 2, 3, 4$, drawn on the floe beside the D1 optima; σ , d_{tot} , κ against n_s (worst case and average).
- Efficiency of the D1 optimum against heading (a polar plot): the cost of placing for the wrong direction.
- The heading-nuisance penalty: $\sigma_{\ln \beta}$ with ψ estimated versus known.
- Revised sensor-count recommendation, and whether it differs from D1's $n_s = 2$.

Trap. Do not average F over headings and then invert: that is the information of an experiment repeated at every heading, which overstates what one deployment at one unknown heading delivers. Average (or take the worst case of) $\text{tr}(F^{-1})$ for D3a, and use the rank-two $\mathbf{S}$, not an average, for D3b. Do not change the frozen estimator or engine: a function that assembles the rank-two $\mathbf{S}$ and passes it to the existing covariance and bias routines is a wrapper, not an engine change; if an engine function cannot accept a general $\mathbf{S}$, record that before adapting anything.

Cost. D1 evaluated 181 104 (layout, reference) pairs per noise case with the vectorised path, and the whole run (three noise cases, gate and Gramian included) took 126 s, so 24 headings and about 7 sea states are minutes to tens of minutes per noise case.

12 M1: Monte Carlo and nonlinear spot checks **MUST** (reduced)

Goal. Show that the analytic predictions hold away from p_0 , where G3 did not test them.

Four scenarios, chosen to stress the linearisation:

1. D1 optimum at $n_s = 2$ (s6, s18; the fewest sensors that should work, as found by D1; rev. 3 had $n_s = 3$ before the result was known);
2. the A1 scenario closest to the Identifiable/Marginal boundary: **miz_R25_h1** ($\sigma_{\ln \beta} = 1.005\%$, Froude-scaled MIZ floe, evaluated in the same scaled setting as A1);
3. R2 stress sea, by the pre-fixed rule **cfg.R2.m1Rule**: $\omega_p = 3.5$ **rad/s**, $\gamma_J = 7$ ($\sigma_{\ln \beta} = 0.766\%$, binding criterion precision);
4. p_0 at LSM6DSV16X noise (the measured-noise twin rerun; an interior point of the R1 window).

Fixed 3 October, all four from their pre-registered rules, before any M1 record is generated. Per scenario: 100 records through the linearised estimator (the standard error of an empirical standard deviation is then about $1/\sqrt{2 \cdot 99} \approx 7\%$, adequate for the acceptance band), extended to 200 only if a result is borderline, (emp/pred and coverage) plus 2 nonlinear inversions from $\pm 20\%$ in β (χ^2/dof , iterations, nonlinear minus linearised). Run overnight in batches.

Pre-registered success criterion. emp/pred in $[0.85, 1.20]$ and 95% coverage in $[0.88, 0.99]$ (the G3.2 gate, relaxed slightly for scenarios far from p_0); nonlinear within 0.2σ of linearised. A failure is reported, not tuned away.

Clarifications logged before the run (3 October; no acceptance value changed). Each case is rebuilt by its source study's code and must reproduce its σ , d_W and class before any record is generated. The linearised estimate and $q = \Delta\theta^T F \Delta\theta$ use the *analytic* J , Σ_k and F (what M1 verifies); q is centred on the truth as in G3a ($E[q] = 2 + d_W^2$). The nonlinear criterion is G3.1's: $|\theta_{\text{NL}} - \theta_{\text{lin}}|/\sigma < 0.2$ per parameter on the same record, starts $\beta \times 0.8$ and $\times 1.2$ with R true. G3's estimator and solver are reused verbatim (`g3EstimatorCore.m`, checked function by function). "Borderline" (extend to 200 records once): the 1-standard-error interval of emp/pred or of the 95% coverage contains an acceptance limit. The mean offset against $\delta\theta_W$ is a diagnostic. Records use the prediction's node grid, so M1 isolates the estimator and the analytic approximation; truth and inverse share the EMM, so model validity stays with V1.

Result (4 October). all four cases pass every criterion (run M1_20261003_221127, commit 96e9b0a, 76 min). emp/pred 0.94 to 1.09; 95% coverage 0.925 to 0.965; mean offsets consistent with $\delta\theta_W$ (within 1.4 SE); all eight nonlinear inversions converged in 3 to 5 iterations and agree with the linearised estimate within 0.018σ , both starts landing on the same point. Weakest case: the Froude-scaled MIZ floe, scatter 8 to 9% above prediction (about 1.5 SE), its A1 class unchanged; quote analytic σ with up to about 10% optimism for floes far from the basin reference. A1 and R2 were extended to 200 records by the borderline rule. `SHARC_M1_Spot_Checks.tex`.

13 H1: calibration and co-located transmissibility **SHOULD**

Goal. Show on real hardware whether the sensors can meet the relative-gain (0.42%) and timing (0.37 ms) tolerances derived in the tolerance study, using the inversion's own estimator. About one day in total, of which about 2 h is logging. Independent of E1, so it can proceed even if the engine slips.

13.1 Why it stays when the array does not

The raw magnitudes $|a| = 10.18, 10.53, 9.93 \text{ ms}^{-2}$ against $g = 9.796$ motivate calibration but do not by themselves measure gain error: they combine offset (datasheet zero- g offsets alone reach 0.39 ms^{-2}), scale, misalignment and non-orthogonality. Without calibration the thesis can only state the 0.42% requirement; with it, the thesis can state whether the requirement is achievable. Offset (bias) itself does not enter the transmissibility: it sits at 0 Hz and is removed by detrending, and the overnight drift (about 1 mg over hours) is far below the band. Bias is reported only for tilt (1 mg is about 0.06°) and as a drift observation.

13.2 H1a: static ellipsoid calibration

- At least 12 (preferably 18) roughly placed stationary orientations including the six faces, each held 2–5 minutes (the Allan minimum of the overnight record: longer averaging is worse).
- Each hold gives *one* equation, $\|\mathbf{a}_{\text{cal}}\| = g$ ($g = 9.796 \text{ ms}^{-2}$, Cape Town), however many samples it contains. The seven holds logged on 30 September therefore support only the **diagonal model** $\mathbf{a}_{\text{cal}} = \text{diag}(s_x, s_y, s_z)(\mathbf{a}_{\text{raw}} - \mathbf{b})$ (6 parameters, 1 degree of freedom); the full model with non-orthogonality (9 parameters) needs at least 10, preferably 12 or more, genuinely 3D holds. Leave-

one-hold-out refits show that the up, down and side-000 holds are each essential, so the geometry is thin.

- **Add three tilted holds** (cylinder leaning at about 45° in a padded box, at three roll angles) if this can be done safely: 10 holds give 4 degrees of freedom and make every leave-one-out refit determined (verified on synthetic data).
- Validate first on synthetic data with known $\mathbf{M}$, $\mathbf{b}$ (`testAccelCalibration.m`).

Outputs: per-sensor scale factors with uncertainties; residual $\|\mathbf{a}_{\text{cal}}\| - g$ RMS; non-orthogonality angles; the relative vertical gain between the three sensors after calibration and its uncertainty, compared with 0.42%.

13.3 H1b: co-located transmissibility (dynamic test)

Three sensors on one rigid board see the same acceleration, so the true transmissibility between any two of them is exactly $T = 1$. Excite the board by hand for about 10 minutes (per orientation used), with energy in 0.5–1.3 Hz, and estimate the transmissibility with the project’s corrected estimator on the vertical channels,

$$\hat{T}_{jr} = \frac{\hat{S}_{jr}}{\hat{S}_{rr} - S_{nn,r}},$$

with $S_{nn,r}$ the reference sensor’s measured floor. Also require excitation strong enough that $S_{nn,r}/\hat{S}_{rr} < 10^{-3}$ in every bin used, so any residual reference-noise attenuation is below 0.1%, a quarter of the tolerance being tested. (Hand excitation of about 1 ms^{-2} rms in the band gives a ratio of order 10^{-5} .) Then

$$|\hat{T}_{jr}| - 1 \Rightarrow \text{relative in-band gain}, \quad \arg \hat{T}_{jr} \approx -\omega \delta t_{jr} \Rightarrow \text{relative delay}.$$

- Before and after applying H1a: the static calibration fixes DC gain; H1b measures the in-band gain, so the difference exposes any internal-filter droop.
- The phase slope measures the combined read-order skew and any difference in internal filter group delay between sensor types (LSM versus ICM). The filter term is not visible in the timestamps and could by itself exceed 0.37 ms, so this is a direct test of the timing tolerance.
- Compare the fitted δt with the logging-path read-order offsets (about 1 ms per sensor), but do *not* subtract one from the other: the logged offset is a logging-path time, and treating it as the physical acquisition offset needs Justin’s confirmation.

Trap. Rotation contaminates the test: sensors a distance d apart see different accelerations $\boldsymbol{\alpha} \times \mathbf{d} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{d})$. With $d \approx 1 \text{ cm}$ and $\alpha \approx 1 \text{ rad s}^{-2}$ that is $\sim 10^{-2} \text{ ms}^{-2}$, a percent-level error on a $\sim 1 \text{ ms}^{-2}$ signal. Excite by vertical translation (board on a flat plate moved up and down, not rocked), check rotation with the gyro, and obtain the sensor positions on the PCB from the layout (Justin) to bound or correct the lever-arm term. Also map each sensor’s axes to the board frame first (sign and axis conventions differ between parts).

13.4 H1c: gravity-on- x hold

One hold of at least 1 h (overnight is fine) with gravity along x , analysed with the static-noise driver (`runStaticNoiseCharacterisation.m`). If the MIZ-band coherence moves to x , that supports the hypothesis of common gain/offset modulation; if it stays on the physical vertical, that supports aliased vibration. Rotating the board does not uniquely identify either mechanism. The first static orientation of H1a can be this hold.

Pre-registered success criterion. the tolerance is demonstrated achievable on this hardware if, after calibration, the band-averaged $|\hat{T}_{jr} - 1|$ and its 95% interval lie within 0.42%, and the fitted $|\delta t_{jr}|$, as measured, and its 95% interval lie within 0.37 ms. If the delay exceeds the tolerance but is repeatable between two runs (within its 95% intervals), it is reported as a correctable deterministic offset, pending Justin’s confirmation of its origin. Otherwise the thesis reports the measured gap as a hardware requirement.

14 Order of operations

14.1 Dependency chain

$$E0 \parallel E1 \longrightarrow \begin{cases} R4, D2, R3 \text{ (quick wins)} \\ D1 \\ A1 \\ R1, R2 \\ V1, V2 \longrightarrow D3b \\ D1 \longrightarrow D3a \end{cases} \longrightarrow M1 \longrightarrow \text{freeze}$$

E0 and H1 are independent of E1 and run in gaps (EMM batches, overnight runs). Nothing except E0 starts before E1 passes its validation gate. M1 needs the D1, A1 and R2 results to choose its scenarios. D3a needs only D1’s tools; D3b needs V2’s forcing model.

14.2 Day plan

Day	Work	Exit condition
Wed 30 Sep (pm)	Pre-register (<code>defineStudyScenarios.m</code>). H1 logging: 12–18 orientations, translation excitation; gravity-on- x hold overnight. Shell-rate check. Message Justin.	Scenarios committed; calibration logs saved.
Thu 1 Oct	E1 build. H1 analysis while EMM solves run. E0 started in gaps.	Σ_k and F at p_0 ; H1 table.
Fri 2 Oct	E1 validation gate (morning). R4, D2, R3. D1. A1 EMM batch overnight. <i>Done: gate and R4 on Thursday; D2, R3, D1 and the A1 batch on Friday.</i>	Gate passed; quick-win tables; D1 figures.
Sat 3 Oct	A1 evaluation and map. R2, R1. Launch M1 overnight. <i>Done: all, M1 launched 22:11.</i>	A1 map; R1/R2 figures; M1 running.
Sun 4 Oct	M1 results. V1, truncation ladder first. E0 start. <i>Done: M1 (all pass) and V1, each with plotter and summary. E0 moved to Tuesday.</i>	All MUST studies done (E0 register Tuesday).
Mon 5 Oct	V2, forcing model first (D3 calls it); then D3 (D3a, then D3b, overnight if needed).	V2 tables; forcing model callable; D3 recommendation.
Tue 6 Oct	E0 register, start and finish. V2 and D3 write-ups. H1 lever-arm correction if PCB positions came late.	Register complete.
Wed 7 Oct	Thesis-grade figures.	Figures final.
Thu 8 Oct	Buffer; stretch items; outline the thesis results chapter.	
Fri 9 Oct	Freeze: every figure and number traceable; one-page results summary.	Experiments stop.

14.3 Slip rules

- If E1 is not validated by Friday evening: drop V1 and V2 entirely, cut A1 to the ring and 6 physical classes, and keep D1, R1, R2 and M1.
- If A1 EMM solves are slower than estimated: run the ring and physical classes only; corners are the first cut.
- V2 and D3 start only if all MUST items are complete (target Sunday 4 October). If only one day remains, keep the V2 forcing model and bias sweep and D3a; reduce D3b to $n_s = 2, 3$ and two sea states.
- H1 logging happens on Wednesday 30 September (it needs the board, not the engine). If Thursday is short, H1 analysis shrinks to H1b before calibration (relative gain and delay as measured) plus H1c; the ellipsoid fit is the cut.
- No study is extended past its day to “make the result nicer”; incomplete sweeps are reported with their coverage.

15 Files

New folder `Inverse Mapping/Parameter Inversion/Design and Robustness/` (next to `Output-Only Twin/`), results in `Inverse Mapping/Parameter Inversion/Results/`.

File	Role	Step
<code>defineStudyScenarios.m</code>	All scenario lists, bands, admissibility rules and class thresholds, fixed before running	Wed
<i>In Design and Robustness/Engine/ (as built, 1 October)</i>		
<code>emmField.m</code>	H and $\partial H/\partial \ln p$ at fixed points (cached node FRF)	E1
<code>transmissibilityFromField.m</code>	T , features and J for any layout and reference	E1
<code>welchBins.m, welchNEff.m</code>	Retained bins and N_{eff} , exactly as the estimator	E1
<code>welchSmoothedSpectrum.m</code>	Window-smoothed spectral matrix	E1
<code>analyticTransmissibilityCovariance.m</code>	Full real Σ_k (Wishart delta method, $N_{\text{eff}} - 1$)	E1
<code>welchExpectedTransmissibility.m</code>	Welch-expected T , hence b_W	E1
<code>fisherFromBlocks.m</code>	F , σ , ρ , κ and bias projection	E1
<code>evaluateScenario.m</code>	Scenario to metric struct and class	E1
<code>Unit Tests/testAnalyticEngine.m</code>	Validation gate against U1, L2048 and G3	E1
<i>In Design and Robustness/ (as built; each study has a run and a plot script)</i>		
<code>Reporting/</code> (14 functions)	Standard style, figure layout and saving, results folders, manifest, CSV; <code>Unit Tests/testReporting.m</code>	Thu
<code>runR4GammaTolerance.m, plotR4GammaTolerance.m</code>	R4	Thu
<code>runD2RecordDuration.m, plotD2RecordDuration.m</code>	D2	Fri
<code>runR3NoiseMisspecification.m, plotR3NoiseMisspecification.m</code>	R3	Fri
<code>d1LayoutTools.m</code>	Grid, enumeration, full-field precompute, vectorised layout search, Gramian ranking (reused by D3)	Fri
<code>runD1SensorLayout.m, plotD1SensorLayout.m</code>	D1 and Gramian comparison; <code>Unit Tests/testD1SensorLayout.m</code>	Fri
<code>a1Tools.m, runA1FloePropertySweep.m, plotA1FloePropertySweep.m</code>	A1: scaling, placement, model checks, checkpointed batch; <code>Unit Tests/testA1FloePropertySweep.m</code>	Fri–Sat
<code>diagnoseTruncationP0.m</code>	A1 pre-flight: truncation ladder at p_0	Fri
<code>buildR2Seas.m, runR2SpectralShape.m, plotR2SpectralShape.m</code>	R2; <code>Unit Tests/testR2SpectralShape.m</code>	Sat
<code>r1Boundary.m, runR1SNR.m, plotR1SNR.m</code>	R1 and boundary bisection; <code>Unit Tests/testR1SNR.m</code>	Sat
<code>g3EstimatorCore.m</code>	G3's estimator and solver, copied verbatim from the G3 script (checked by the tester)	Sat
<code>resolveM1Scenarios.m, m1Tools.m</code>	M1 cases rebuilt by their source studies' code; records, batches, checkpoints, solver wiring	Sat
<code>runM1MonteCarloSpotChecks.m, plotM1MonteCarloSpotChecks.m</code>	M1; <code>Unit Tests/testM1MonteCarloSpotChecks.m</code> (12 checks)	Sat–Sun
<code>diagnoseV1Truncation.m</code>	V1 stage 0: truncation ladder at p_0 , large- kR last step, P check, V1a preview	Sun
<code>v1Tools.m, runV1ModelMismatch.m, plotV1ModelMismatch.m</code>	V1: truth models, Welch features, projection, tolerances, nonlinear check; <code>Unit Tests/testV1ModelMismatch.m</code> (11 checks)	Sun
<i>In Forward Model/</i>		
<code>dispersionRoots.m</code>	Damped-root fallback (only when the seed grid finds nothing); <code>Unit-Tests/testDampedRootFallback.m</code>	Sat
<i>Planned</i>		
<code>directionalSpectralMatrix.m</code>	Rank- n spectral matrix for directional forcing (shared by V2 and D3)	Mon
<code>runDirectionalForcingStudy.m</code>	V2	Mon
<code>runD3DirectionalLayout.m, plotD3DirectionalLayout.m</code>	D3a and D3b, reusing <code>d1LayoutTools.m</code>	Tue
<i>In IMU Characterisation/Processing/</i>		
<code>calibrateAccelSixPosition.m</code>	H1a ellipsoid fit, $N \geq 12$ orientations (name kept from the IMU plan)	Thu
<code>Unit-Tests/testAccelCalibration.m</code>	Known $\mathbf{M}$, $\mathbf{b}$ recovery on synthetic orientations	Thu
<code>runColocatedTransmissibility.m</code>	H1b gain and delay between co-located sensors, before/after calibration	Thu
<code>Unit-Tests/testColocatedTransmissibility.m</code>	Synthetic board with known gain (1.003) and delay (1 ms)	Thu

16 What the thesis gains

- **Design rules:** how many sensors, where (for a known heading and, from D3, for an unknown heading and a directional sea), which reference, how long to record, how well γ , the noise floor, the water depth and the sensor geometry must be known, and which EMM truncation real data need. Together with the Monday tolerances, a complete requirement set for the deferred array.
- **A domain of applicability:** which floe classes (basin, tank, MIZ) are identifiable, organised by non-dimensional groups.
- **Verified analytic machinery:** Monte Carlo and nonlinear spot checks (M1) confirm the analytic Fisher and Welch-bias predictions in the hardest design, domain, forcing and noise cases.
- **Robustness envelope:** measured sensors placed in an SNR window bounded by precision on one side and estimator bias on the other, sea-shape sensitivity, and the distinction between estimator error and forward-model discrepancy (V1: truncation and sensor geometry matter, depth does not).
- **A hardware anchor:** whether real sensors meet the gain and timing tolerances after calibration, measured with the inversion's own estimator (H1).
- **A validity diagnostic (V2):** whether coherence warns that the single-input assumption has failed.

Narrative. Validated hydroelastic model $\rightarrow$ SSI investigated systematically $\rightarrow$ poles unsuitable as the primary observable $\rightarrow$ output-only transmissibilities $\rightarrow$ finite-record estimator validated $\rightarrow$ inverse closed (G3) $\rightarrow$ real IMU noise characterised $\rightarrow$ **design, robustness and domain of the complete method** (this programme).